

JYOTIRADITYA SINGH

Software Engineer & Data Engineer

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Innovative Software & Data Engineer leveraging a robust foundation in Python, SQL, and system architecture to build scalable solutions. Adept at designing ETL data pipelines, deploying machine learning inference models, and optimizing full-stack backend services. Passionate about writing clean, efficient code and integrating Generative AI capabilities to drive measurable business value.

TECHNICAL SKILLS

Programming Languages:	Python, SQL, JavaScript, C/C++
Data Engineering & AI:	ETL Pipelines, Data Analytics, Machine Learning, Generative AI (LLMs, APIs), Pandas, NumPy
Backend & Databases:	Node.js, RESTful APIs, MySQL, MongoDB, Redis
Developer Tools:	Git, GitHub, VS Code, Postman, Selenium, Agile, MATLAB

PROFESSIONAL EXPERIENCE

DRDO (ISSA Department) — Research Intern, Data & Systems Architecture 06/2025 – 07/2025

Location: New Delhi, India

- Architected data evaluation frameworks to process and analyze large-scale Electronic Warfare (EW) datasets, executing capability-gap analyses across multi-branch defense systems.
- Evaluated and benchmarked machine learning algorithms against legacy hardware constraints, proposing software-defined upgrade architectures that enhanced computational efficiency.
- Authored comprehensive technical documentation (30+ pages) detailing system architectures, API integration strategies, and data flow models to streamline cross-functional workflows.

PROJECTS

AI-Based Multilingual Application Backend 01/2025 – 05/2025

Tech Stack: Node.js, MongoDB, Redis, Python, Generative AI APIs

- Engineered a highly scalable marketplace backend leveraging Node.js, MongoDB, and Redis caching, facilitating low-latency data retrieval and concurrent user sessions.
- Deployed Python-based machine learning inference pipelines (Pandas, NumPy) and integrated Generative AI APIs to automate real-time plant disease classification and dynamic advisory generation.
- Optimized backend RESTful API endpoints and data processing pipelines, reducing ML model response times to sub-500 milliseconds and significantly improving user experience.

IoT Telemetry & Analytics Pipeline Self-Initiated

Tech Stack: Python, Pandas, NumPy, ESP8266/NodeMCU, IoT

- Architected an end-to-end IoT telemetry pipeline utilizing ESP8266 microcontrollers to ingest real-time environmental data streams into a centralized database.
- Authored robust Python scripts for automated ETL (Extract, Transform, Load) processes, utilizing Pandas to clean anomalous sensor data and execute event-driven alerts.

EDUCATION

SRM Institute of Science and Technology 09/2022 – 05/2026

B.Tech in Electronics & Communication Engineering | CGPA: 7.81/10.00 | Chennai, India

- Coursework:** Data Structures & Algorithms, DBMS, Object-Oriented Programming, Operating Systems, Computer Networks

D.A.V. Public School Delhi, India

- Class XII (CBSE) — 86.3% • Class X (CBSE) — 80.5%

CERTIFICATES & VIRTUAL INTERNSHIPS

- Software Engineering Bootcamp:** Python Pro Developer
- Microsoft & AICTE Virtual Internship:** Cloud & AI Technologies
- Google & AICTE Virtual Internship:** Android Developer Program